Cluster : Computer & Network Security

Problem Statement title : Cloud Misconfiguration (Security Scanner)

Description:

Develop a tool to audit cloud infrastructure configurations and detect security misconfigurations across AWS, Azure, and GCP environments, enabling early identification of risky exposures.

Exact Deliverables :

- Rules engine for cloud configuration checks (aligned with CIS benchmarks and IaC tools like Terraform).

- Automated detection of common misconfigurations (e.g., open storage buckets, insecure security groups).

- Validation tests by spawning misconfigurations and measuring time-to-detect vs. baseline scanners.

- Dashboard to visualize findings and generate compliance reports.

Milestones, Evolution Parameters:

- Phase 1: Build a basic rules engine for misconfig detection.

- Phase 2: Integrate with IaC pipelines and validate against seeded misconfig libraries.

- Phase 3: Develop visualization and compliance reporting features.

- KPIs: Detection coverage, false positives, detection latency, compliance readiness.

Additional Information:

- Focus on scalability to multi-cloud environments.

- Ensure the scanner is modular, supporting integration with CI/CD pipelines.

- Students should study major breaches caused by cloud misconfigs to design realistic test cases.